PAPER ANALYSIS

From foundations to current research in representation learning

Four sessions, five papers per session, group presentations, and joint synthesis

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Four Paper Analysis Sessions

Session	Topic	Focus
1 · Nov 27	Implicit Neural Representations and Approximation	Continuous representations, function approximation, coordinate networks, Fourier features, neural fields
2 · Dec 4	Location Encoders	Geographic coordinates, spatial inductive biases, spherical geometry, learned location representations
3 · Dec 11	Self-Supervised Representation Learning	Contrastive learning, self-distillation, masked reconstruction, multimodal learning
4 · Dec 18	Geospatial Embeddings	Earth representations, explicit and implicit embeddings, global embedding systems, embedding spaces

Progression: INRs → Location Encoders → Self-Supervised Learning → Geospatial Embeddings

Your Paper Presentation

Each group presents one paper for approximately **20 minutes**. Explain the paper rather than only summarizing it.

1. **Problem and motivation:** What problem does the paper address, and why does it matter?
2. **Main idea:** What is the central contribution? Explain it in your own words.
3. **Technical method:** Explain the key mechanism, representation, architecture, or mathematical idea. Focus on why it works.
4. **Evidence and results:** Which experiments support the claims? What are the most important results?
5. **Limitations:** What does the method leave unresolved? Which assumptions or weaknesses matter?
6. **Technical impact:** What did the paper introduce, popularize, clarify, or enable? How does it connect to the other papers?
7. **Two exam questions:** Write open-ended questions with answer sketches that test concepts rather than facts.

> **Prefer "Why does the method need this representation, and what would likely happen without it?" over "Which dataset did the authors use?"**

Paper Analysis Session Timeline

Time	Activity
10:00–10:10	Topic framing and introduction
10:10–10:40	Group 1 presentation and discussion
10:40–11:10	Group 2 presentation and discussion
11:10–11:40	Group 3 presentation and discussion
11:40–12:00	Cross-paper discussion and buffer
12:00–13:00	Break
13:00–13:30	Group 4 presentation and discussion
13:30–14:00	Group 5 presentation and discussion
14:00–14:50	Synthesis across all five papers: shared findings, differences, and impact on the field
14:50–15:00	Wrap-up and assignment of the next session's papers

Session 1: Implicit Neural Representations

November 27

How can neural networks represent continuous signals, functions, and scenes?

1. **Review:** Neural Fields in Visual Computing and Beyond
   Xie, Takikawa, Saito, Litany et al. *Computer Graphics Forum* 41(2), 2022.

2. **SIREN:** Implicit Neural Representations with Periodic Activation Functions
   Sitzmann, Martel, Bergman, Lindell, Wetzstein. *NeurIPS 2020*.

3. **Fourier Features:** Fourier Features Let Networks Learn High Frequency Functions in Low Dimensional Domains
   Tancik, Srinivasan, Mildenhall, Fridovich-Keil et al. *NeurIPS 2020*.

4. **NeRF:** NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis
   Mildenhall, Srinivasan, Tancik, Barron, Ramamoorthi, Ng. *ECCV 2020*.

5. **Generalisation:** Generalised Implicit Neural Representations
   Grattarola, Vandergheynst. 2022.

Synthesis: representation, frequency, architecture, application, and generalization

Session 2: Location Encoders

December 4

How should a machine-learning model represent a location on Earth?

1. **Review:** A Review of Location Encoding for GeoAI: Methods and Applications
   Mai, Janowicz, Hu, Gao et al. *International Journal of Geographical Information Science* 36(4), 2022.

2. **Spherical Harmonics and SIREN:** Geographic Location Encoding with Spherical Harmonics and Sinusoidal Representation Networks
   Rußwurm, Klemmer, Rolf, Zbinden, Tuia. *ICLR 2024*.

3. **Spatial INR:** Spatial Implicit Neural Representations for Global-Scale Species Mapping
   Cole, Van Horn, Lange, Shepard et al. *ICML 2023*, PMLR 202:6320–6342.

4. **Spherical Attention:** Attention on the Sphere
   Bonev, Rietmann, Paris, Carpentieri, Kurth. *NeurIPS 2025*.

5. **To be announced:** Paper 5

Synthesis: Where should spatial geometry enter the model: encoding, representation, or architecture?

Session 3: Self-Supervised Learning

December 11

How can useful representations emerge without task-specific labels?

1. **SimCLR:** A Simple Framework for Contrastive Learning of Visual Representations
   Chen, Kornblith, Norouzi, Hinton. *ICML 2020*, PMLR 119:1597–1607.

2. **DINO:** Emerging Properties in Self-Supervised Vision Transformers
   Caron, Touvron, Misra, Jégou et al. *ICCV 2021*, pp. 9650–9660.

3. **MAE:** Masked Autoencoders Are Scalable Vision Learners
   He, Chen, Xie, Li, Dollár, Girshick. *CVPR 2022*, pp. 16000–16009.

4. **CLIP:** Learning Transferable Visual Models From Natural Language Supervision
   Radford, Kim, Hallacy, Ramesh et al. *ICML 2021*, PMLR 139:8748–8763.

5. **Analysis:** What Do Self-Supervised Vision Transformers Learn?
   Park, Kim, Heo, Kim, Yun. *ICLR 2023*.

Synthesis: contrastive learning, self-distillation, masked reconstruction, multimodal learning, and representation analysis

Session 4: Geospatial Embeddings

December 18

Can information about the Earth be compressed into reusable embedding spaces?

1. **Perspective:** Earth Embeddings: Toward Artificial Intelligence-Centric Representations of Our Planet
   Klemmer, Rolf, Rußwurm, Camps-Valls et al. *IEEE Geoscience and Remote Sensing Magazine*, 2026.

2. **AlphaEarth:** AlphaEarth Foundations: An Embedding Field Model for Accurate and Efficient Global Mapping from Sparse Label Data
   Brown, Kazmierski, Pasquarella et al. 2025.

3. **Analysis:** Measuring the Intrinsic Dimension of Earth Representations
   Rao, Rußwurm, Klemmer, Rolf. 2025/2026.

4. **SatCLIP:** SatCLIP: Global, General-Purpose Location Embeddings with Satellite Imagery
   Klemmer, Rolf, Robinson, Mackey, Rußwurm. *AAAI 2025*, 39(4):4347–4355.

5. **TESSERA:** TESSERA: Temporal Embeddings of Surface Spectra for Earth Representation and Analysis
   Feng, Atzberger, Jaffer, Knezevic et al. *CVPR 2026*.

Synthesis: What is an Earth embedding, what information does it contain, and how should we evaluate it?